Competition under Interdependence

Exchange institutions and scarce coordination in a Catan-inspired resource economy

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RESEARCH QUESTION

When agents are resource-interdependent but compete over development, which kinds of exchange institutions sustain output?

INTERDEPENDENCE

Complementary bundles

Development projects require different combinations of materials, components, food, energy, and knowledge.

BEHAVIOR

Different objectives

Agents use weighted decision rules that differ in capacity focus, reserve protection, competition, cooperation, and fairness.

INSTITUTIONS

Exchange is organized

Bargaining, pooling, redistribution, and central matching preserve or override different behavioral hooks.

Agent Construction

$$\mathrm{Agent}_i = \begin{cases} x_i & \text{resource stock} \\ a_i & \text{built assets and production capacity} \\ s_i & \text{development score} \\ p_i & \text{asymmetric access profile} \end{cases}$$

$$p_i = (0.45_{\text{primary}},\ 0.25_{\text{secondary}},\ 0.10,\ 0.10,\ 0.10)$$

Individually incomplete resource access

Each agent has strong access to two inputs and only weak access to the remaining three. Across the five-agent population, primary and secondary access are balanced.

The economy contains all required inputs, but individual agents repeatedly encounter resource bottlenecks and depend on exchange.

Round structure

1 Produce

Agents draw resources from asymmetric access profiles. Production sites and advanced sites increase future draws.

2 Exchange

The selected institution can bargain, pool, subsidize, or centrally match resources before building starts.

3 Build

Agents sequentially spend bundles on up to four projects per round, so one exchange can unlock several investments.

4 Update

Scores, production capacity, access bonuses, and leader crowns affect later bottlenecks and exchange positions.

Build rules and capacity

PROJECT	COST	DIRECT EFFECT	DYNAMIC ROLE
Infrastructure	**1M + 1C**	0 points	two units create one site slot
Production site	**1M + 1C + 1F + 1E**	+3 points	adds one draw and improves weakest access weight
Advanced site	**2E + 3K**	+5 points	upgrades one production site and adds further production
Innovation	**1F + 1E + 1K**	+1 point for first three	counts toward the innovation crown

$$\text{free capacity}_i = \left\lfloor \frac{\text{infrastructure}_i}{2} \right\rfloor - (\text{production sites}_i + \text{advanced sites}_i)$$

Behavioral profiles are weighted decision rules

$$V_i(b \mid x_i) = \underbrace{w_{\text{points}} \Delta s_i(b) + w_{\text{innovation}} I(b) + w_{\text{crown}} K_i(b)}_{\text{immediate scoring motives}}$$
$$+ \underbrace{w_{\text{capacity}} \Delta C_i(b) + w_{\text{site}} \Delta P_i(b)}_{\text{productive development}} - \underbrace{w_{\text{progress}} M_i(b \mid x_i)}_{\text{distance from completion}}$$

b = candidate build project

x_i = agent i's current stock and development state

Every project receives a value from this expression. The highest-valued project becomes the agent's current target.

From build targets to exchange decisions

1 Identify the bottleneck

$$m_{ir} = \max\{0,\ \text{target requirement}_{ir} - \text{current stock}_{ir}\}$$

The agent requests up to two units of the resource with the largest remaining shortfall.

2 Protect reserves

Expendable surplus = stock − target cost − profile reserve

Only positive surplus can be offered in bargaining or contributed to a public pool.

3 Evaluate the proposed transfer

$$\Delta W_i(T) = W_i(\text{stock after } T) - W_i(\text{stock before } T)$$

Offer

An agent proposes terms when the it gains enough through an offer

Accept

Accept when $\Delta W_i(T) \geq \tau_i^{\text{accept}}$

Contribute

Pool surplus when the profile's contribution rule permits it.

Behavioral profiles differ through weights

$$\theta_i = \left(w_{\text{own}}, w_{\text{capacity}}, w_{\text{site}}, w_{\text{social}}, w_{\text{rival}}, w_{\text{equity}}, w_{\text{security}}, \tau_i^{\text{offer}}, \tau_i^{\text{accept}}\right)$$

All agents use the same target-selection and transfer-evaluation rules. Behavioral profiles change the weights, protected reserve margin and minimum utility changes required to offer or accept.

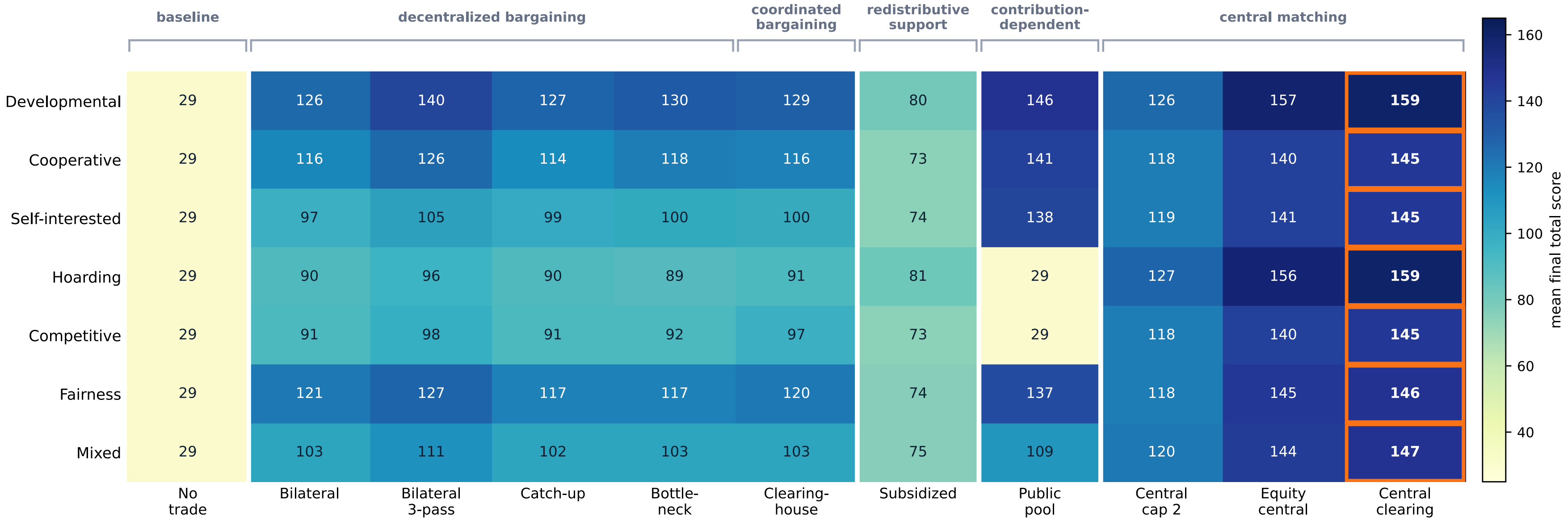
PROFILE	W_{OWN}	W_{CAPACITY}	W_{SITE}	W_{SOCIAL}	W_{RIVAL}	W_{EQUITY}	W_{SECURITY}
Developmental	1.5	2.0	2.4	0.2	0.2	0.0	0.25
Cooperative	1.0	1.8	2.0	1.0	0.0	0.2	0.15
Self-interested	1.6	1.8	2.0	0.0	0.3	0.0	0.35
Security / hoarding	1.1	1.5	1.7	0.0	0.1	0.0	1.00
Competitive	1.5	1.7	1.9	−0.1	1.0	0.0	0.40
Fairness-oriented	1.0	1.7	1.8	0.4	0.0	1.0	0.25

Eleven institutions vary who controls exchange

FAMILY	CONDITION	EXCHANGE PHASE EFFECT
Baseline	**No trade**	Resources remain with their producers. Agents can only build from their own stocks.
Decentralized bargaining	**Bilateral**	Basic trading protocol. Requester executes cheapest acceptable offer. Comes in 3 variants that modify trading passes and request ordering.
	Bilateral 3-pass	**The request–offer–accept procedure repeats up to three times per round using updated stocks.**
	Catch-up priority	**Requests from lower-scoring agents are processed first.**
	Bottleneck priority	**Requests closest to completing current build target are processed first.**
Contribution-dependent	**Public pool**	Agents voluntarily contribute surplus. Institution allocates pooled resources to target shortfalls.
Central matching	**Equity central**	Central matching prioritizing swaps for lower-scoring requesters.
	Central clearing	**Repeatedly constructs and executes target-safe swaps until no beneficial matches remain.**

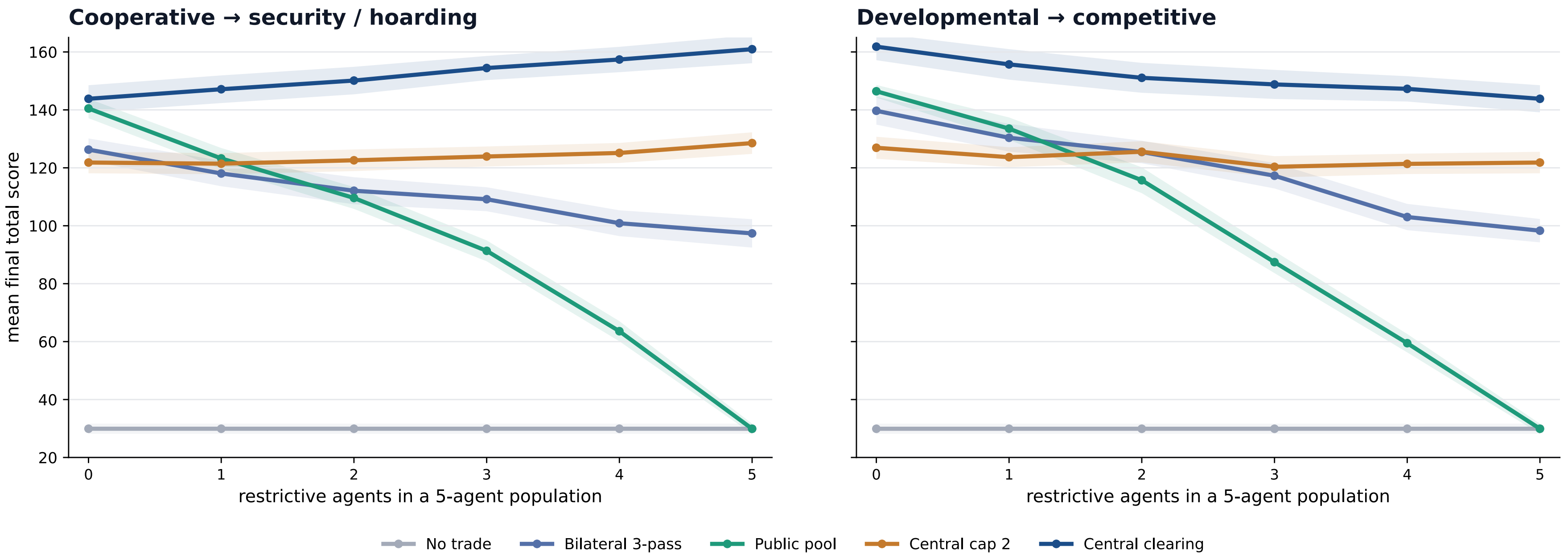
Free central matching dominates the unconstrained benchmark

Development-oriented build mode; mean final total score over 200 matched seeds. Orange outline marks the row winner.



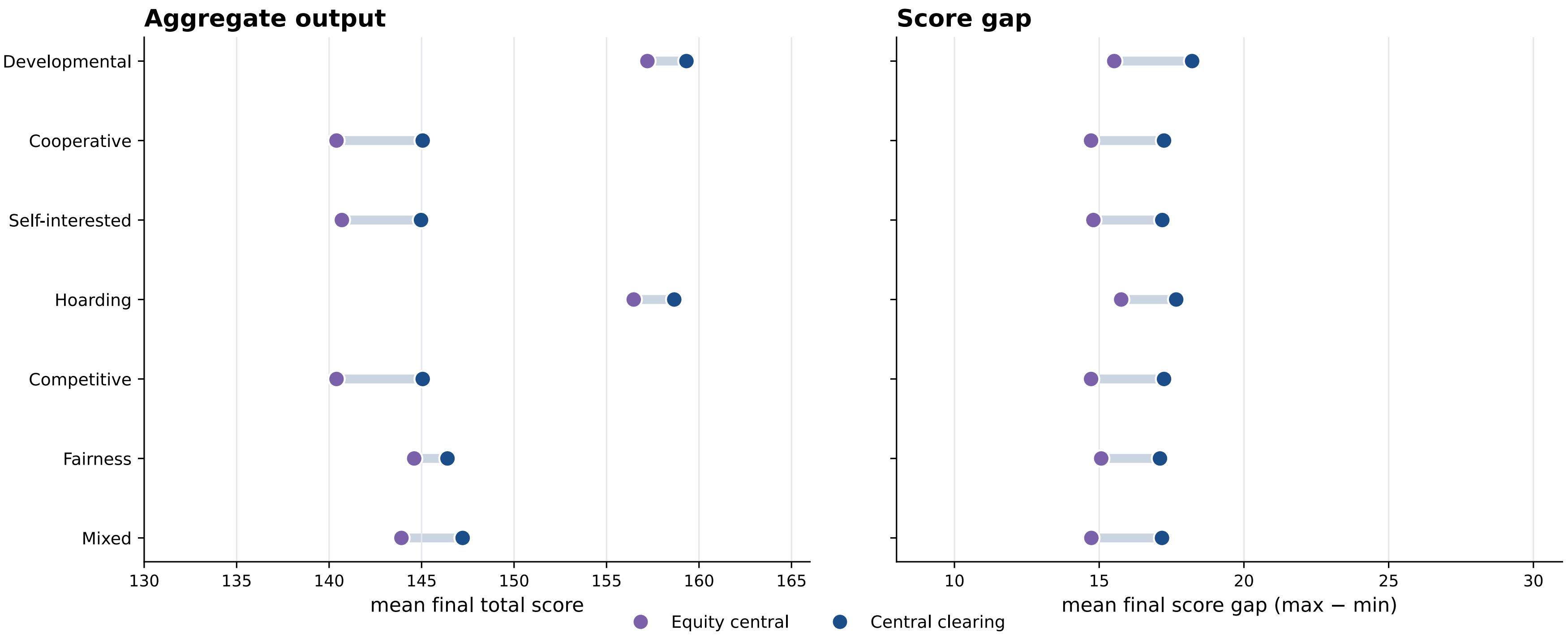
Behavioral composition changes which institution works

Population composition sweeps; 100 matched seeds per cell.



Equity weighting trades some output for smaller score gaps

Central clearing versus equity-weighted central matching by population; lower gap is more equal development.



Making central coordination scarce

Coordinated institutions require governance capacity to set up and operate

The realized cost of institution a in round t is:

$$c_t(a) = b(a) + s_t(a) + w_t(a)$$

$b(a)$	**Fixed operating burden:** Between 0 and 3 with Bilateral 3-pass being 0 and full Central Clearing being 3.
$s_t(a)$	**Switching cost:** 1 unit when entering a different non-bilateral institution. Free when retaining the same institution or returning to bilateral.
$w_t(a)$	**Variable workload cost:** A bounded capacity charge based on the resource units processed by the institution.

How capacity constrains choice

Before selection, institution a is available only if capacity covers its maximum possible cost:

$$k_t(a) = b(a) + s_t(a) + w^{\max}(a) \leq G_t$$

After operation, only the realized cost is deducted and one unit regenerates:

$$G_{t+1} = \min\{G_{\max},\, G_t - c_t(a_t) + g\}$$

Spending capacity now can restrict which institutions remain feasible later.

Capacity mechanics & affordability

Example: Full Central Clearing

Maximum commitment when switching in:
3 fixed + 1 switching + 2 maximum workload = **6**

Realized operation:
Two reciprocal trades process four resource units. Workload cost is 1 as long as processed units are less than 7. Workload cost is:
3 fixed + 1 switching + 1 realized workload = **5**

Capacity update:
$G_t = 8$
$G_{t+1} = 8 - 5 + 1 = 4$

Maximum costs per institution:

- Bilateral 3-pass **(0)**
- Clearinghouse **(3)**, Subsidized catch-up **(3)**
- Round-local public pool **(4)**, Central capped (2) **(4)**, Equity central capped (2) **(4)**
- Full central clearing **(6)**

Q-learning planner

1 State

The planner maps the post-production game state into seven discretized features: capacity, round phase, score inequality, development bottlenecks, idle infrastructure, voluntary exchange viability, and the previous institution.

- **Exchange viability:** share of requests that can receive acceptable, feasible offers.

2 Action

It chooses among institutional families:

- **Voluntary exchange:** Bilateral 3-pass, Clearinghouse
- **Collective / targeted support:** Round-local public pool, Subsidized catch-up
- **Central matching:** Central capped (2), Equity central capped (2), Full central clearing

3 Reward

$$\Phi_t = (1-\lambda)\bar{s}_t + \lambda s_t^{\min}$$

$$r_t = \Phi_{t+1} - \Phi_t$$

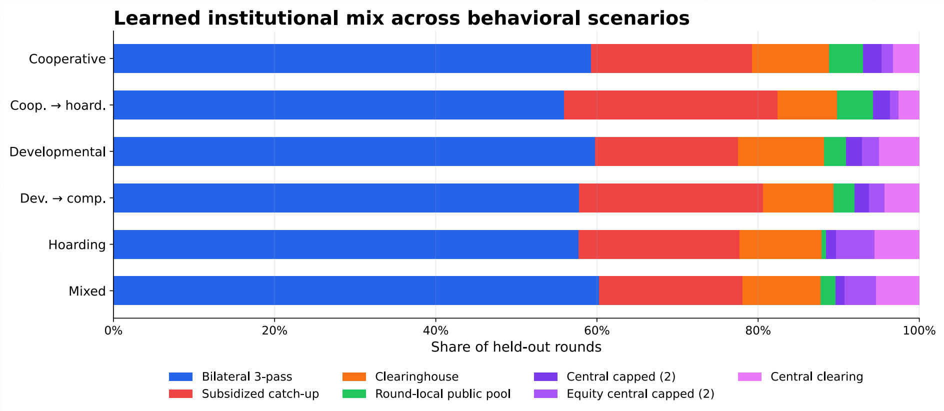
- $\bar{s}_t$: mean score across the five agents
- s_t^{min} : weakest-agent score
- λ = 0.25 : weight on weakest-agent outcome

With λ = 0.25,the weakest-agent outcome receives 25% welfare weight.

Training: 12.000 episodes. The planner explores random available institutions, but increasingly selects the highest-valued action as ε falls from 1.0 to 0.03.

Evaluation: frozen Q-table on 100 held-out seeds per scenario; exploration and learning are switched off.

The trained policy produces a mixed regime



Outcomes

58.5% Bilateral 3-pass remains dominant.

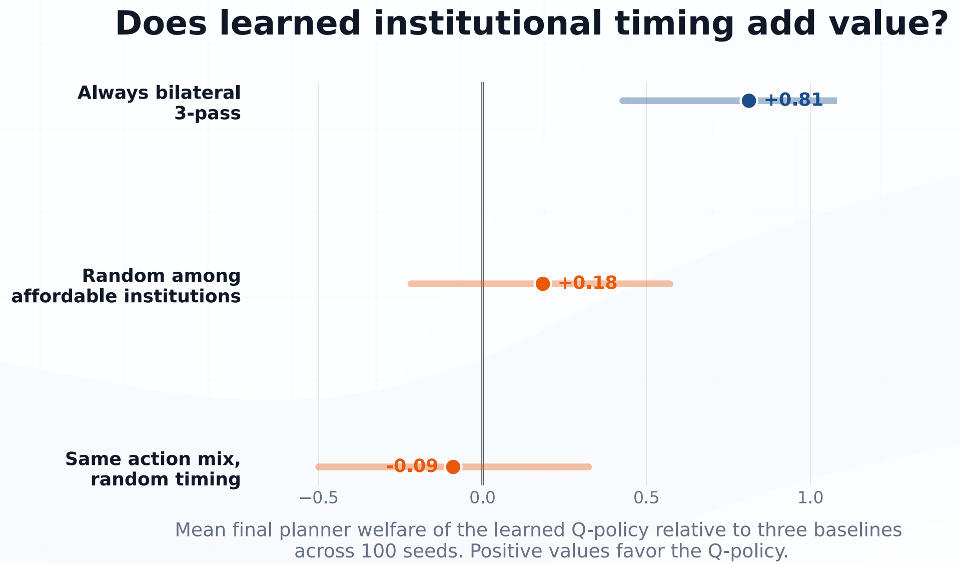
20.8% Subsidized catch-up supplies targeted lower-tail support.

9.4% Clearinghouse adds coordination while retaining voluntary offers.

4.3% Full central clearing is used selectively.

The capacity mechanism prevents permanent centralization. The open question is whether the Q-table learns **when** to use interventions.

The learned timing does not pass the baseline comparisons



Interpretation of Baselines

- **Random affordable:** random choice among institutions allowed by current capacity.
- **Same mix, random timing:** preserves the learned policy's overall action frequencies but removes state-dependent timing.

Result: The Q-policy does not reliably outperform random affordable selection or the same institutional mix with randomized timing.

Controlled rollouts show a learnable signal

Pilot diagnostic: from 36 checkpoints across six base games, i cloned the complete state, tried every feasible next institution, and then let the retained Q-policy control the remaining rounds.

Choices at the checkpoints affected final welfare

The Monte Carlo rollout-estimated best institution exceeded the runner-up by 1.43 final planner-welfare points on average.

The Q-policy rarely matched the rollout leader

The institution selected by the retained Q-policy matched the rollout-estimated best checkpoint action in only 6 of 36 sampled decisions.

Estimated loss from the Q-policy's choice

Keeping the Q-policy for all later decisions, its selected institution finished 3.12 final welfare points below the rollout-estimated best alternative on average.

Better institutional choices appear to exist at specific decision points, but the tabular Q-policy does not identify them reliably. The result was not unique to Q-policy continuation: with random feasible choices afterward, the estimated top-two gap remained 0.96 welfare points.

Conclusion

A modular model for resource exchanges

The model separates agent behavior from institutional organization, allowing the same economy to be tested under different behavioral populations and coordination mechanisms.

Governance as a scarce resource

Once institutions consume limited governance capacity, permanent central coordination is no longer the obvious solution. The planner must decide when intervention is worth sacrificing future flexibility.

Good interventions are hard to identify

Controlled rollouts suggest that different institutions are preferable in specific situations and at different points in time. The tested Q-learners, however, did not identify those situations reliably.

The model provides a controlled way to study which institutions can improve outcomes, as well as what kind of information an adaptive planner needs to choose between them at the right time.